

Class 2 Assignment – DAX Fundamentals (Financials Sample)

Part 1 – Create Basic Measures

Create these measures:

```
Total Sales = SUM(Financials[Sales])
Total Profit = SUM(Financials[Profit])
Total COGS = SUM(Financials[COGS])
Total Gross Sales = SUM(Financials[Gross Sales])
Total Discount = SUM(Financials[Discounts])
Units Sold = SUM(Financials[Units Sold])
```

Part 2 – Average Measures

Create:

```
Average Sales = AVERAGE(Financials[Sales])
Average Profit = AVERAGE(Financials[Profit])
Average Unit Price = AVERAGE(Financials[Sale Price])
```

Part 3 – Maximum & Minimum

```
Highest Sale = MAX(Financials[Sales])
Lowest Sale = MIN(Financials[Sales])
Highest Profit = MAX(Financials[Profit])
Lowest Profit = MIN(Financials[Profit])
```

Part 4 – Count Functions

```
Total Records = COUNTROWS(Financials)
Total Countries = DISTINCTCOUNT(Financials[Country])
Total Products = DISTINCTCOUNT(Financials[Product])
Total Segments = DISTINCTCOUNT(Financials[Segment])
```

Part 5 – DIVIDE Function

```
Profit % =
DIVIDE(
    [Total Profit],
    [Total Sales],
```

```
0
)
Discount % =
DIVIDE(
    [Total Discount],
    [Total Gross Sales],
    0
)
```

Part 6 – Calculated Columns

Sales Category

```
Sales Category =
IF(
    Financials[Sales] >= 1000000,
    "High",
    "Low"
)
```

Profit Category

```
Profit Category =
SWITCH(
    TRUE(),
    Financials[Profit] >= 300000, "Excellent",
    Financials[Profit] >= 150000, "Good",
    Financials[Profit] >= 50000, "Average",
    "Poor"
)
```

Month Short Name

```
Month Short =
LEFT(Financials[Month Name],3)
```

Year Month

```
Year Month =
FORMAT(Financials[Date], "MMM YYYY")
```

Part 7 – Text Functions

Create:

```
Country Length =  
LEN(Financials[Country])  
First 3 Letters =  
LEFT(Financials[Country],3)  
Last 2 Letters =  
RIGHT(Financials[Country],2)
```

Part 8 – Date Functions

```
Year = YEAR(Financials[Date])  
Month = MONTH(Financials[Date])  
Day = DAY(Financials[Date])
```

Part 9 – Variables (VAR)

```
Profit Margin =  
VAR SalesValue = [Total Sales]  
VAR ProfitValue = [Total Profit]
```

```
RETURN  
DIVIDE(  
    ProfitValue,  
    SalesValue,  
    0  
)
```

Part 10 – Cards

Create Card visuals for:

- Total Sales
 - Total Profit
 - Total COGS
 - Gross Sales
 - Discount
 - Profit %
 - Units Sold
-

Part 11 – Charts

Create:

1. Clustered Column Chart

- Axis: Country
 - Values: Total Sales
 - 2. Bar Chart
 - Axis: Product
 - Values: Total Profit
 - 3. Pie Chart
 - Legend: Segment
 - Values: Total Sales
 - 4. Line Chart
 - Axis: Month Name
 - Values: Total Sales
 - 5. Matrix
 - Rows: Country
 - Columns: Product
 - Values: Total Sales
-

Part 12 – Slicers

Add slicers for:

- Year
 - Country
 - Product
 - Segment
 - Discount Band
-

Final Dashboard

Create a one-page dashboard with:

- 7 KPI Cards
- 4 Charts
- 1 Matrix
- 5 Slicers
- Proper titles and formatting

Assignment Challenge (Advanced)

Without looking at notes, create these measures:

- Total Sales
- Total Profit
- Profit %
- Average Sales
- Total Countries
- Total Products
- Highest Sale
- Lowest Sale
- Discount %
- Profit Margin (using `VAR`)